Airline Passenger Satisfaction Analysis

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Executive Summary

This report analyzes the Airline Passenger Satisfaction dataset to understand the main factors that affect customer satisfaction in an airline company. The dataset includes passenger information, travel class, flight distance, delay times, service ratings, and overall satisfaction results.

The main objective of this analysis is to identify why many passengers are neutral or dissatisfied and to provide useful recommendations for management. Python was used for the analysis, including Pandas for data cleaning and analysis, NumPy for numerical operations, and Matplotlib for visualizations.

The analysis shows that only 43.3% of passengers are satisfied, while the majority are neutral or dissatisfied. The most important factors related to satisfaction are service quality variables, especially Online Boarding, Inflight Entertainment, and Seat Comfort. Business Class passengers are much more satisfied than Economy and Eco Plus passengers, while delays have a weaker but still negative effect on satisfaction.

Overall, the airline should focus on improving service quality, especially in Economy class, while also protecting its strong Business Class performance and reducing operational delays.

Dataset Description

The dataset for this project is taken from Airline Passenger Satisfaction dataset on Kaggle. It includes information on passengers, flight details, ratings of service quality, delay times and customer satisfaction.

Each row of the data set represents one passenger's flight experience. The analysis is based on a cleaned dataset of 103594 passengers. This dataset is appropriate for business analysis as it has both numerical and categorical variables related to customer experience and airline operations.

The main categorical variables are Gender, Customer type, Type of Travel, Class and satisfaction. These variables allow comparisons of satisfaction between different passenger groups.

The key numerical features are Age, Flight Distance, Departure Delay in Minutes, Arrival Delay in Minutes and the service rating columns Online Boarding, Seat Comfort, Inflight Entertainment, Checkin Service, Inflight Service and Cleanliness.

The main target variable is satisfaction which indicates whether a passenger is satisfied or neutral or dissatisfied. This variable is utilized to comprehend which factors are most associated with positive or negative passenger experience.

The dataset was chosen because it has enough rows, numerical columns, and categorical columns to perform meaningful business data analysis.

Methodology & Analysis

The analysis was completed using Python. First, the dataset was imported with Pandas using pd.read_csv(). The structure of the dataset was checked with functions such as head(), info(), and isnull().sum().

Before the analysis, data cleaning steps were applied. The unnecessary index column was removed because it did not provide useful information for the analysis. Missing values were checked, and rows with missing values in Arrival Delay in Minutes were removed. Duplicate rows were also checked.

After cleaning the data, new variables were created to support the analysis. Satisfied Flag was created by using np.where(), where satisfied passengers were coded as 1 and neutral or dissatisfied passengers were coded as 0. This made it possible to calculate satisfaction rates numerically.

Delay Category was created to group arrival delays into categories such as no delay, small delay, medium delay, and long delay. Distance Category was created to classify flights as short, medium, or long distance. Average Service Score was calculated by taking the average of several service-related columns. This variable summarizes the overall service quality experienced by each passenger.

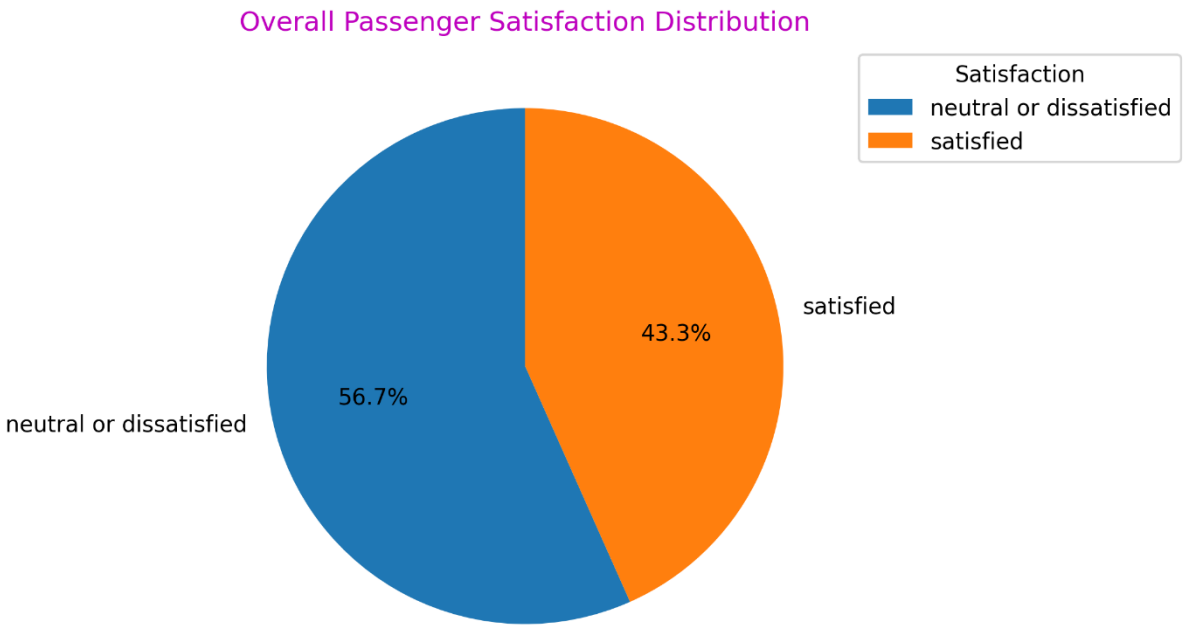
Several Pandas methods were used in the analysis. groupby() was used to compare satisfaction rates by travel class, delay category, distance category, and age group. pd.crosstab() was used to compare categorical variables such as distance category and travel class. corr() function was used to create a correlation matrix and understand the relationship between numerical variables.

Matplotlib was used to create all charts in the report. These visualizations helped identify the main patterns in the dataset and supported business recommendations. The analysis focuses not only on showing charts, but also on explaining what decisions management should make based on the results.

Visualizations & Interpretation

Overall Passenger Satisfaction Distribution

The pie chart below shows the baseline satisfaction level across the entire dataset of 103,594 passengers.



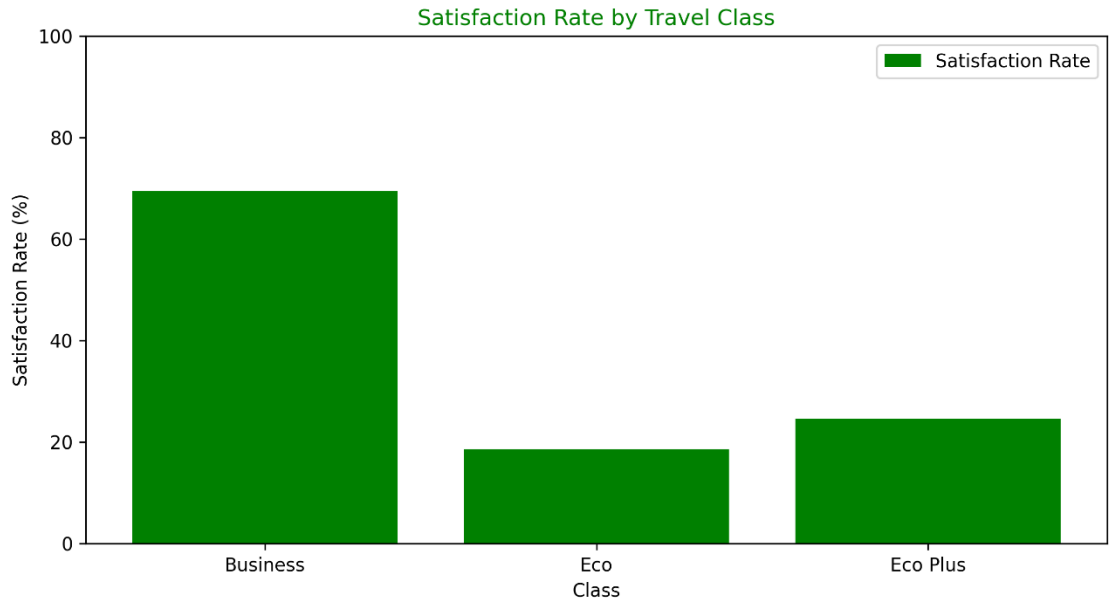
A majority of passengers are neutral or dissatisfied, while only **43.3%** are satisfied. This means that less than half of the passengers had a positive experience. For an airline company, this is an important problem because customer satisfaction affects loyalty and future travel choices.

This chart shows the main business issue of the project: the company needs to understand why many passengers are not satisfied.

The next analyses focus on the possible reasons behind this result.

Satisfaction Rate by Travel Class

The bar chart below compares satisfaction rates across Business, Eco Plus, and Economy cabin classes.



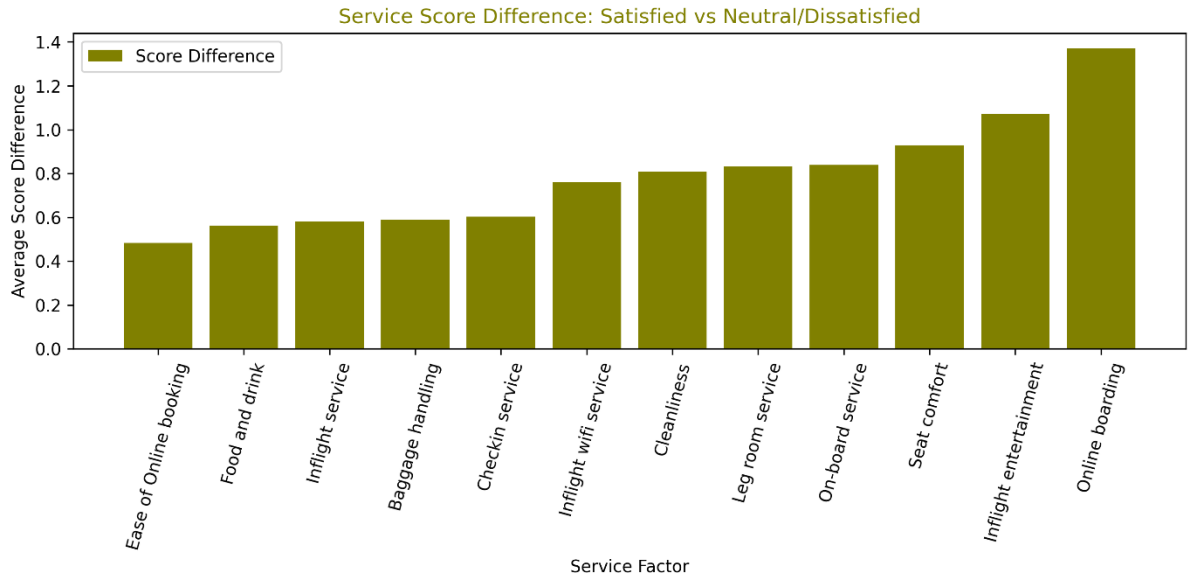
Business Class passengers have a **69.4%** satisfaction rate. This is much higher than Economy passengers, with **18.6%,** and Eco Plus passengers, with **24.6%.** This is one of the clearest findings in the dataset.

The result shows that Business Class passengers are much more satisfied because they usually receive better comfort, entertainment, legroom, and service. On the other hand, Economy and Eco Plus passengers have low satisfaction rates. This means Eco Plus may not be different enough from regular Economy to improve the passenger experience.

The airline should focus on improving the Economy cabin because most passengers travel in this class. If the company does not improve Economy service quality, the overall satisfaction rate will probably remain low.

Service Score Difference

The chart compares average service scores between satisfied and neutral/dissatisfied passengers.



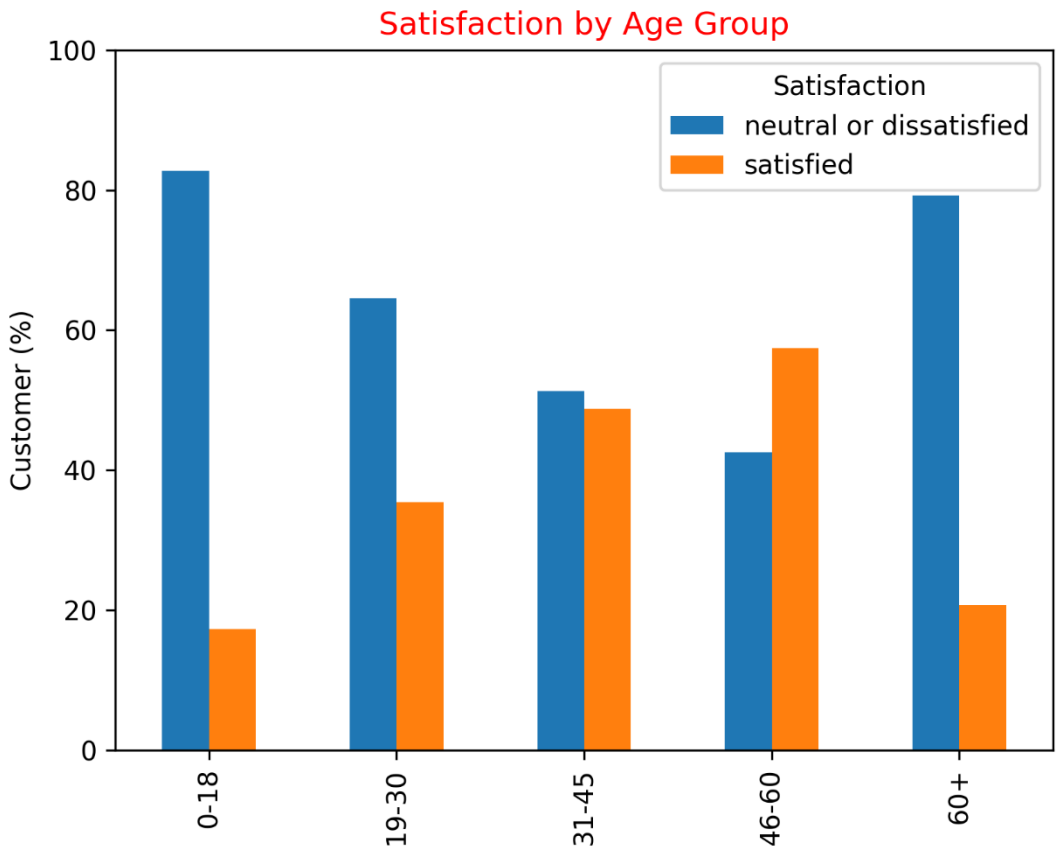
Online Boarding has the largest difference, with **1.37** rating points. It is followed by Inflight Entertainment with **1.07** points and Seat Comfort with **0.93** points. These results show that these three service areas are strongly related to passenger satisfaction.

On the other hand, Ease of Online Booking, Food and Drink, and Inflight Service have smaller differences. This means satisfied and dissatisfied passengers gave more similar ratings in these areas.

This chart is very useful for management because it clearly shows which service areas should be improved first. The airline should focus mainly on Online Boarding, Inflight Entertainment, and Seat Comfort to increase passenger satisfaction.

Satisfaction by Age Group

The grouped bar chart below breaks down satisfaction and dissatisfaction rates across five passenger age cohorts.

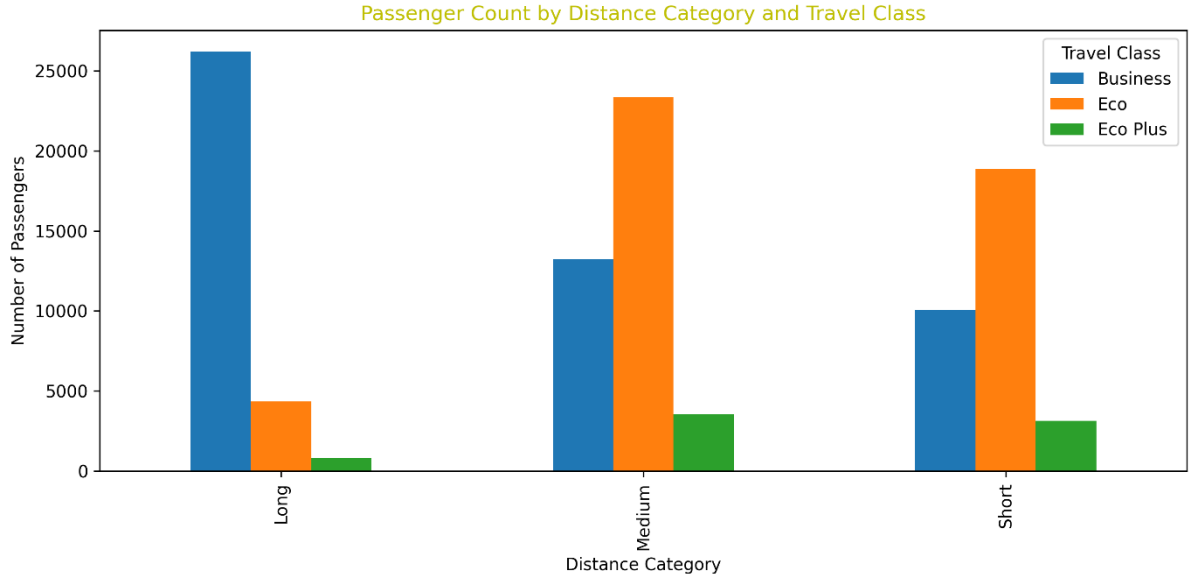


The age group chart shows that satisfaction is different across passenger groups. The **46–60** age group has the highest satisfaction rate, while the **0–18** and **60+** groups have higher neutral or dissatisfied rates.

This result shows that passengers from different age groups may have different expectations. Younger passengers may expect better digital services, such as online boarding and Wi-Fi. Older passengers may care more about comfort, support, and reliable service. Therefore, the airline should not use the same service strategy for every age group.

Passenger Count by Distance Category and Travel Class

The chart shows the number of passengers by flight distance category and travel class. The results show that passenger distribution changes clearly depending on distance.



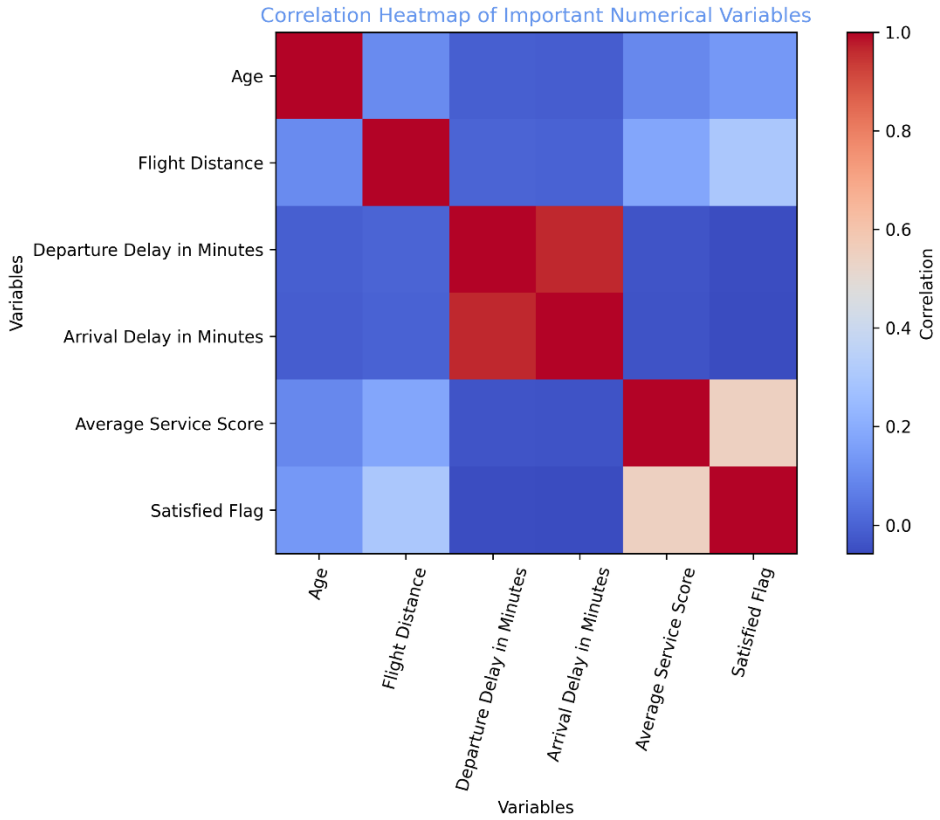
The main finding is that Business Class is the largest group in long-distance flights with 26,231 passengers. However, for short and medium-distance flights, Economy passengers are the largest group.

This means passenger profiles change depending on flight distance. Long-distance flights are more important for Business Class, while short and medium routes depend more on Economy passengers.

The airline should protect its strong Business Class performance on long-distance flights. At the same time, it can improve Business Class packages for short and medium-distance flights with better comfort, priority boarding, and flexible services to increase demand in these segments.

Correlation Heatmap of Key Numerical Variables

The correlation heatmap shows relationships between important numerical variables.



Average Service Score positively correlates with Satisfied Flag, higher service score means passengers are more likely to be satisfied.

On the other hand, there are weak negative relations between satisfaction and Departure Delay and Arrival Delay. Delays decrease satisfaction, but not as strongly as service quality.

The airline should focus to improve the service quality. But minimizing delays should also remain an important operational objective.

Key Insights & Actionable Recommendations

The analysis shows that the airline has a low overall satisfaction rate. Only **43.3%** of passengers are satisfied, which means the company should improve its customer experience.

The most important factor related to satisfaction is service quality. Online Boarding, Inflight Entertainment, and Seat Comfort have the largest differences between satisfied and dissatisfied passengers. Therefore, these areas should be the first improvement priorities.

Travel class is also an important factor. Business Class passengers are much more satisfied than Economy and Eco Plus passengers. However, Economy passengers represent a large customer group, especially in short and medium-distance flights. Improving Economy service quality can help increase the overall satisfaction rate.

The analysis also shows that Business Class is strong in long-distance flights. At the same time, there is potential to improve Business Class offers in short and medium-distance flights with better comfort, priority boarding, and flexible services.

Delays have a negative effect on satisfaction, but service quality has a stronger relationship with satisfaction. Because of this, the airline should mainly focus on improving service quality while also continuing to reduce delays.

Recommendations:

- Improve Online Boarding, Inflight Entertainment, and Seat Comfort.
- Improve Economy service quality, especially on short and medium-distance flights.
- Protect strong Business Class performance on long-distance flights.
- Develop better Business Class offers for short and medium-distance routes.
- Continue reducing departure and arrival delays.

Conclusion

This project indicates that the airline has a serious satisfaction problem as only 43.3% of the passengers are satisfied. Service quality is the major factor associated with satisfaction, especially Online Boarding, Inflight Entertainment and Seat Comfort.

The correlation heatmap also supports this result, as it shows that Average Service Score has a positive relationship with satisfaction. Delays reduce satisfaction, but their effect is less strong than service quality.

Overall, the airline needs to improve key service areas, protect its strong Business Class performance on long-haul flights, and develop better Business Class offer for short and medium distance routes. Such actions can help to improve customer satisfaction and to strengthen the competitive position of the airline.

References

Kaggle. Airline Passenger Satisfaction Dataset: https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction